

SNAPFRAME · REQUIREMENTS

SnapFrame requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/snapframe		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	Apply SNF-DDR-001 (R10 redefined as the frame kit; 1 in rafters); status from SNF-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

SnapFrame requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be revised after co-design sessions. The status column now comes from the TRL 3 calculation note SNF-CAL-001, for the size M frame with 1 in EMT rafters and 3/4 in EMT elsewhere (SNF-DDR-001). Five requirements are **not met**: R5 (packages), R6 (wind), R7 (skin), R10 (cost) and R13 (fire). Three are at risk: R8 (anchors), R9 (nodes) and R12 (part count).

ID	Requirement	Target	Status (SNF-CAL-001)	Verification (TRL 3 or later)
R1	Give enough covered living space	Size M: 16 m ² floor or more, which is 3.5 m ² per person for four people (Sphere minimum)	Met: 4.0 x 4.0 m = 16.0 m ²	Geometry
R2	Give standing headroom	2.0 m or more over at least 60 % of the floor	Met: 73 % of the floor, measured to the rafter underside	Geometry
R3	Cover several sizes from one design	At least three sizes (S about 9 m ² , M 16 m ² , L about 24 m ²) generated from one parametric node model; M and L share every tube length	Met: <code>cad/src/model.py</code> generates and exports the S, M and L node sets	Parametric model
R4	Go up without tools	Two adults with no tools and no prior training, using a picture guide, erect frame and skin of size M in 60 min or less	Not verifiable at TRL 3: estimate about 46 min	Time study in a later user trial (TRL 4 or later)
R5	Ship flat	No member longer than 2.1 m; all members straight; kit in two packages, each 25 kg (55 lb) or less and 0.10 m ³ or less	Not met: longest member 2.064 m, but the tube bundle weighs about 27.4 kg; bag 22.4 kg with tarpaulins	Massing model, then weighing
R6	Resist wind with the skin on	Every member stays elastic with a safety factor of 1.5 or more on yield in gusts of 20 m/s (72 km/h, 45 mph)	Not met: rating about 17.8 m/s (64 km/h, 40 mph); 3/4 in ridge tubes reach 1.18 and posts 1.46 at 20 m/s; 1 in rafters reach 1.66	Wind load calculation (CAL), then frame analysis
R7	Fit standard relief tarpaulins	Two 4 x 6 m tarpaulins (48 m ²) close the roof, both side walls and both gables	Not met: full enclosure needs about 51.4 m ² ; two tarpaulins close the roof, both side walls and one gable. How to treat the open gable is proposed, awaiting Amish	Cutting and folding plan
R8	Anchor without a hammer	Every foot and guy anchored by hand; each anchor holds 1.0 kN uplift in firm soil	At risk: estimated demand up to 1.28 kN at a rear corner foot before cable pretension; holding capacity not verifiable at TRL 3	Anchor pull-out data for the chosen anchor and soils
R9	Keep nodes strong in sun and cold	Nodes carry the socket forces from R6 with a factor of 2 after two years outdoors, from -10 to 70 °C surface temperature	At risk: hitch pin bearing on the socket about 32 MPa with the factor of 2; polymer not chosen	Material data, then node load tests (TRL 4)
R10	Stay within the concept budget	Size M frame kit \$400 or less in prototype quantities, excluding tarpaulins, which come from agency stock and are costed separately (SNF-DDR-001 D1)	Not met: frame kit about \$431 (7.8 % over); tarpaulins about \$50 extra	Priced BOM
R11	Be carried by two people	Complete size M kit, including tarpaulins, 50 kg (110 lb) or less	Met: frame kit about 40.6 kg; 49.8 kg with tarpaulins	Massing model, then weighing
R12				Design review

ID	Requirement	Target	Status (SNF-CAL-001)	Verification (TRL 3 or later)
	Be repairable in the field	Any member or node replaced by hand in 5 min or less; four tube types and three node types only	At risk: four tube types; three node families, but six printed variants because corner eave nodes are handed	
R13	Limit fire spread	Skin meets a recognized flame spread test for tents (for example CPAI-84) or the kit ships with a fire-retardant tarpaulin option	Not met with standard polyethylene relief tarpaulins, which are usually not flame retardant	Material data for the chosen tarpaulin

■ Assumptions

- Reference size M (SNF-DDR-001 D6): 4.0 x 4.0 m floor, two 2.0 m bays, eave 1.8 m, ridge 2.6 m (roof pitch 21.8°), three gable frames.
- Tube (SNF-DDR-001 D2): 1 in EMT rafters (OD 29.54 mm, wall 1.448 mm, 1.003 kg/m) and 3/4 in EMT elsewhere (OD 23.42 mm, wall 1.245 mm, 0.681 kg/m), ANSI C80.3 nominal dimensions. Yield strength assumed 275 MPa, to be confirmed from supplier data.
- Wind: $q = \frac{1}{2}\rho v^2$ with $\rho = 1.225 \text{ kg/m}^3$; windward wall +0.8, leeward wall -0.5, roof suction -0.7 (first-order values in the style of ASCE/SEI 7, not a code check). Each tarpaulin panel shares its load between its four edge members by 45° tributary lines. See SNF-CAL-001 for the full list.
- Members are treated as simply supported between node centers; node fixity is ignored, which is conservative for bending.
- Snow and cyclone rating are out of scope for the first release (SNF-DDR-001 D7).
- Budget: \$400 is the `budget_usd` in `project.yaml`, applied to one size M frame kit including anchors, cables and bags, and excluding tarpaulins (SNF-DDR-001 D1).